Jaccard and Dice Dissimilarity

2026-04-17

Introduction

Jaccard and Dice dissimilarities are the standard distance measures for binary (presence/absence) data: ecological community composition, disease symptom panels, gene set membership, document term frequencies after thresholding. Both indices share a key property: they ignore joint absences and focus only on agreements and disagreements among presences. This is exactly what is needed when an absence carries little information — a species absent from two sites might be absent because the habitat is unsuitable, or simply not detected — but is wrong when absences are themselves diagnostic.

Prerequisites

A working understanding of binary data, set-based similarity reasoning, and the conventional 2×2 contingency table for two binary vectors.

Theory

For two binary vectors x, y with counts a (both present), b (only in x), c (only in y), d (both absent):

- **Jaccard index**: $J = a/(a+b+c)$, the ratio of the intersection to the union of the present-set. **Jaccard dissimilarity** is $1 - J$.
- **Dice / Sørensen coefficient**: $D = 2a/(2a+b+c)$, with **Dice dissimilarity** $1 - D$. Dice doubles the weight on joint presences relative to mismatches.
- **Sørensen** and **Dice** are equivalent for binary data.

Both ignore d (joint absences), making them appropriate when absences are uninformative. Indices that include d — simple matching, Hamming distance — are preferable when joint absences carry meaning (e.g., binary medical diagnoses where "neither has condition X" is informative).

Assumptions

Binary data (or count data thresholded to binary), and joint absences are not meaningful for the inferential question. The same data on different absence-meaning assumptions can yield very different distances.

R Implementation

```r
library(vegan)

# Species abundance data (reduce to presence/absence)
data(dune)
dune_pa <- (dune > 0) * 1
head(dune_pa)

# Jaccard dissimilarity
d_jac <- vegdist(dune_pa, method = "jaccard", binary = TRUE)
as.matrix(d_jac)[1:3, 1:3]
```

```r
# Sorensen (= Dice)
d_sor <- vegdist(dune_pa, method = "bray", binary = TRUE)
```

Output & Results

Two pairwise dissimilarity matrices on the dune presence/absence data, one for Jaccard and one for Sørensen-Dice. Jaccard distances are systematically larger than Dice for the same data because Dice double-counts joint presences in its denominator.

Interpretation

A reporting sentence: "Jaccard dissimilarity averaged 0.40 across all pairs of sites (range 0.16 to 0.65), indicating that 40 % of species presences are not shared between any randomly chosen pair; the corresponding Sørensen-Dice dissimilarity averaged 0.25, reflecting Dice's higher weight on joint presences." Always state which index was used; the same data produce different conclusions under Jaccard versus Dice.

Practical Tips

- Use Jaccard or Dice when joint absences are biologically or ecologically uninformative (rare species, sparse term-document data, gene-set membership).
- For abundance rather than presence-absence data, Bray-Curtis is the standard; Bray-Curtis on binarised data reduces to Sørensen-Dice.
- For microbiome data with phylogenetic information, UniFrac (`GUniFrac::GUniFrac`) is preferable because it weights distances by evolutionary relationships, not just shared OTUs.
- For binary medical data where joint absences are meaningful (neither patient has the condition), use simple matching (`dist(x, method = "binary")` for Hamming-style) rather than Jaccard.
- For very large sparse binary matrices (text, single-cell sparse counts), use the sparse-matrix-aware implementations in `proxy` or `parallelDist` to avoid memory blow-up.
- Pair dissimilarity matrices with NMDS or PCoA for visualisation; both ordinations consume Jaccard or Dice matrices directly.

R Packages Used

`vegan` for `vegdist()` with `method = "jaccard"` or `"bray"` and the broader ecology workflow; `proxy` for general dissimilarity functions including binary metrics; `philentropy` for distance and similarity coefficients across many families; `GUniFrac` for phylogenetic UniFrac alternatives in microbiome analysis.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE